Re-Evaluating Dual Use Technology Regulation in the Age of AI

Arik Karim

Introduction

Artificial intelligence (AI) is the defining technological breakthrough of the past decade. Far removed from its theoretical beginnings in the 1950s, AI's horizontal, general-purpose characteristics lend themselves to transformative applications across scientific, educational, and economic domains, among others. Yet, these same qualities render AI a paradigmatic dual-use technology, capable of advancing the public good while simultaneously posing national security risks. While AI's applications are easily cognized at a macro-level—chatbots that democratize information access, agents that automate tasks, and proprietary tools used for scientific discovery—there are five key characteristics that distinguish AI meaningfully from other technologies.

First, AI is a general-purpose technology, capable of learning tasks across many domains without being designed for any single application. Unlike specialized tools that serve discrete functions, AI systems can be trained and retrained for purposes well beyond their original design. Second, AI's operational needs and infrastructure costs are incredibly low and declining by approximately 30% year after year, making advanced capabilities increasingly accessible to state and non-state actors alike.[1] Third, AI tools are inherently software-based and thus intangible, existing as code and data rather than physical objects that can be inspected at borders or tracked through traditional supply chains. Fourth, once developed, AI systems are easily scalable: a model trained at great expense can be quickly copied and deployed millions of times at marginal cost.[2] Finally, AI systems' internal

[1] Stanford Inst. for Human-Centered Artificial Intelligence, *2025 AI Index Report* 69 (2025), https://hai.stanford.edu/assets/files/hai_ai_index_report_2025.pdf.
[2] Ege Erdil et al., *GATE: An Integrated Assessment Model for AI Automation* 13 (Epoch AI 2025), https://arxiv.org/pdf/2503.04941.

decision-making processes operate as a "black box," making the monitoring of intent, end use, and treaty compliance extraordinarily difficult even when access to the system is available.[3]

These characteristics coalesce into a dual-use dilemma with unique challenges for existing regulatory frameworks. The tension between civilian benefit and military risk has long characterized emerging technologies. Chemical compounds developed for agriculture have been weaponized; lasers designed for medical procedures have been adapted for military targeting; uranium enrichment serves both energy production and nuclear weapons development; and drones transition seamlessly between commercial applications and battlefield surveillance. History demonstrates that acting upon dual-use dilemmas early is vital for two distinct reasons.

First, the foreseeability of dual-use dilemmas has reshaped the trajectory of entire technological fields. Nuclear technology's evolution is instructive. Following the devastation of Hiroshima and Nagasaki, the international community recognized that the technology's development was inevitable. This realization led to the creation of the 1953 Atoms for Peace program and the establishment of the International Atomic Energy Association (IAEA), institutions designed to close information asymmetries and verify treaty compliance white promoting peaceful applications.[4] Rather than attempting to suppress nuclear technology entirely, these frameworks sought to channel development toward beneficial ends while establishing mechanisms to detect military diversions.[5] Second, early action upon dual-use dilemmas sparks dialogue within developing technology communities, providing feedback that inform policymakers before norms and practices become internalized. The bioethics community's response to the mousepox virus experiments in the 2000s illustrates this dynamic. When Australian researchers inadvertently created a lethal mousepox strain while attempting to develop a contraceptive vaccine for pest control, the scientific community immediately recognized the implications

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[3] Yavar Bathaee, *The Artificial Intelligence Black Box and the Failure of Intent and Causation*, 31 Harv. J.L. & Tech. 889, 901 (2018), https://jolt.law.harvard.edu/assets/articlePDFs/v31/The-Artificial-Intelligence-Black-Box-and-the-Failure-of-Intent-and-Causation-Yavar-Bathaee.pdf.

[4] Elisa D. Harris (ed.), *Governance of Dual-Use Technologies: Theory and Practice*, ch. 1 – "On the Regulation of Dual-Use Nuclear Technology," in *Governance of Dual-Use Technologies: Theory and Practice* (Am. Acad. Arts & Scis. 2016), Section 4, https://www.amacad.org/publication/governance-dual-use-technologies-theory-and-practice/section/4.

[5] 1995 Review and Extension Conference of the Parties to the Treaty on the Non-Proliferation of Nuclear Weapons, 13th Mtg., U.N. Doc. NPT/CONF.1995/SR.13, at 2 (Apr. 27, 1995), https://docs.un.org/en/NPT/CONF.1995/SR.13.

for bioweapons development.[6] Rather than waiting for government mandates, researchers initiated discussions about publication ethics, laboratory security protocols, and the responsibility of scientists working with dual-use biological research. This self-regulatory impulse, reinforced by early policy intervention, established norms that continue to shape synthetic biology research today. The international response to dual-use technologies primarily manifests itself through the creation of (1) regulatory institutions, such as the IAEA, and (2) multilateral export control regimes, voluntary, informal agreements that coordinate national export controls on dual-use items among supplier countries.

Understanding how to regulate AI requires understanding how AI is built. There are three components that form the so-called "triad" of AI development: data, algorithms, and compute.[7] Data serve as the training material that drives the machine learning process for an AI model, providing the examples from which patterns are extracted and capabilities emerge. Algorithms constitute the core instruction set that informs the compute how to learn from the data, recognize patterns, and operate autonomously—the mathematical and logical frameworks that transform raw information into functional intelligence. Compute refers to the hardware used to provide key resources such as processing power, memory, and storage capacity for AI systems to function.

Among these three pillars, compute stands apart as the most legally tractable chokepoint for regulation. Unlike data, which exists in countless distributed copies across the internet and can be scraped, synthesized, or generated, and unlike algorithms, which spread through academic publication and open-source repositories the moment they are discovered, compute hardware remains a rivalrous physical good. Chips must be manufactured in specialized facilities, transported through identifiable supply chains, installed in specific data centers, purchased by legally registered entities, and powered by measurable electricity.[8] Each advanced processor carries a serial number; each data center occupies a geographic location subject to national jurisdiction; each training run consumes quantifiable resources that generate heat, require cooling, and leave digital traces.

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[6] Seumas Miller & Michael J. Selgelid, *Ethical and Philosophical Consideration of the Dual-Use Dilemma in the Biological Sciences*, 13 SCI. & ENG'G ETHICS 523, 554 (2007), https://pmc.ncbi.nlm.nih.gov/articles/PMC7089176/.
[7] Ben Buchanan, *The AI Triad and What It Means for National Security Strategy* 1 (Ctr. for Security & Emerging Tech., Aug. 2020), https://cset.georgetown.edu/wp-content/uploads/CSET-AI-Triad-Report.pdf.
[8] Erich Grunewald & Christopher Phenicie, *How AI Chips Are Made*, Inst. for AI Policy & Strategy (Sept. 4, 2025), https://www.iaps.ai/research/how-ai-chips-are-made.

Compute is uniquely detectable, excludable, and quantifiable in ways that data and algorithms are not.[9]

This physical reality creates regulatory leverage. A nation cannot easily prevent adversaries from accessing training data or reading published papers describing state-of-the-art architectures, but it can control who manufactures advanced chips, where those chips are shipped, and who is granted access to the computational infrastructure those chips enable. The concentration of chip manufacturing capacity in a handful of companies—NVIDIA, Taiwan Semiconductor Manufacturing Company (TSMC), Samsung, Intel, and Advanced Micro Devices—and the even greater concentration of the equipment required to manufacture cutting-edge chips in firms like Advanced Semiconductor Materials Lithography (ASML), which holds a near-monopoly on extreme ultraviolet lithography systems, create natural chokepoints where regulatory intervention can be effective.[10] Similarly, cloud computing services, despite their apparent diffusion across global networks, ultimately occur in data centers owned by identifiable corporations subject to legal jurisdiction and regulatory pressure.

The nascent international legal frameworks currently governing AI indicate a conspicuous gap in addressing national security applications. The European Union (EU)'s AI Act of 2024, the most comprehensive AI regulation enacted to date, establishes risk-based requirements for AI systems deployed within EU markets.[11] Yet the Act explicitly carves out national security uses from its regulatory scope, deferring to member states' sovereign prerogatives in defense and intelligence matters.[12] Similarly, the Council of Europe's Framework Convention on AI of 2024, which represents the first international treaty specifically addressing artificial intelligence, includes provisions allowing states to exempt AI systems used for national security purposes from the Convention's requirements regarding human rights protection and democratic oversight.[13]

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[9] Girish Sastry et al., *Computing Power and the Governance of Artificial Intelligence* 24 (Centre for the Governance of AI, Feb. 14 2024), https://cdn.governance.ai/Computing_Power_and_the_Governance_of_AI.pdf.
[10] *Id.* at 31.
[11] Rosamund Powell, *The EU AI Act: National Security Implications*, CETaS Explainer 2 (Jul. 31, 2024), https://cetas.turing.ac.uk/sites/default/files/2024-07/cetas_explainer_-_the_eu_ai_act_-_national_security_implications.pdf.
[12] *Id.* at 5.
[13] Council of Europe, *The Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law*, https://www.coe.int/en/web/artificial-intelligence/the-framework-convention-on-artificial-intelligence.

These carveouts are neither accidental nor surprising. They exhibit the push-and-pull between international cooperation on emerging technologies and states' unwillingness to limit their own defense capabilities. Ergo, if the most advanced regulatory frameworks explicitly decline to govern military AI applications, and if traditional export control regimes prove ill-suited to AI's unique qualities, then AI's dual-use dilemma certainly unfolds without effective international constraint.

The issue is therefore not whether AI should be regulated as a dual-use technology—the national security stakes are far too great to ignore—but rather how existing legal frameworks must be recalibrated to address AI's distinctive characteristics. The answer requires understanding both the evolution of dual-use technology regulation and the specific ways in which AI's diffusion mechanisms render traditional export controls inadequate. Part I of this Note will contextualize the regulatory methods available in the American legal toolkit, describe the rise and fall of multilateral export control policymaking, and examine the succeeding shift toward unilateral export controls in the context of dual-use technologies. Part II will discuss existing enforcement gaps in AI, as well as its implications on export control evasion. Finally, Part III will offer a three-tiered compute governance framework centered on controlling access, capability, and post-deployment behavior for the United States. In doing so, it charts a legal path forward to rectify national security concerns with the realities of disaggregated regulations.

I. The Evolution of Dual-Use Technological Regulation

A. Description of Regulatory Methods

Export controls, defined as a set of federal regulations that restrict the release of certain items, information, and software to foreign nationals both within the country and abroad, are the common mechanism of dual-use technological regulation.[14] These controls operate primarily at the point of transfer, requiring licenses for specified transactions and empowering customs authorities to interdict prohibited shipments. Generally, export controls rely heavily on classification systems, such as the United States Commerce Control List (CCL) or the EU Common Military List, that enumerate controlled items

[14] 15 C.F.R. § 730 et seq. (2012) (Export Administration Regulations), https://www.govinfo.gov/content/pkg/CFR-2012-title15-vol2/pdf/CFR-2012-title15-vol2-subtitleB-chapVII-subchapC.pdf.

with varying degrees of specificity.[15] The legal architecture assumes that controlled items can be identified, that transfers occur as discrete events subject to regulatory scrutiny, and that violators can be sanctioned through denial of export privileges or criminal prosecution.

Inbound investment and foreign direct investment screening represent a complementary approach that regulates not the export of technology but rather the acquisition of companies and assets by foreign entities.[16] These mechanisms provide a process through which national governments review incoming foreign investments to determine their potential impact on national security and potentially block their execution. The Committee on Foreign Investment in the United States (CFIUS), the EU's foreign direct investment screening framework, and similar bodies in other jurisdictions evaluate whether foreign acquisitions of domestic companies would provide adversaries with access to sensitive technologies, critical infrastructure, or proprietary information that could be leveraged for military purposes.[17] This mechanism has become increasingly important as instances of illegal direct technology transfers have emerged, discussed in Part II of this Note.

Intellectual property, contracts, and licensing controls encompass a set of legal and commercial mechanisms used to govern the transfer, use, and restriction of sensitive or proprietary technologies through private law instruments. Governments can mandate that certain technologies developed with public funding or classified as sensitive include contractual restrictions on downstream use, transfer, or disclosure.[18] Patent secrecy orders allow governments to prevent publication of innovations deemed to threaten national security.[19] While these mechanisms lack the coercive force of export controls backed by

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15 U.S. DEP'T OF COM., INT'L TRADE ADMIN., *How Do I Determine My Export Control Classification Number (ECCN)*, https://www.trade.gov/how-do-i-determine-my-export-control-classification-number-eccn.
16 Andrea Viski, *Foreign Direct Investment (FDI) Screening: A Primer* 10 (Stimson Center, June 2024), https://www.stimson.org/wp-content/uploads/2024/06/Foreign-Direct-Investment-FDI-Screening-A-Primer.pdf.
17 U.S. DEP'T OF THE TREASURY, *The Committee on Foreign Investment in the United States (CFIUS)*, https://home.treasury.gov/policy-issues/international/the-committee-on-foreign-investment-in-the-united-states-cfius.
18 David Fletcher, *US Government Rights in Technical Data, Software and Subject Inventions: A Primer for Early-Stage, Dual-Use Companies*, Cooley GO (Feb. 11, 2026), https://www.cooleygo.com/us-government-rights-in-technical-data-software-and-subject-inventions-a-primer-for-early-stage-dual-use-companies/.
19 35 U.S.C. § 181 (2024).

criminal sanctions, they create legal obligations enforceable through civil litigation and provide contractual frameworks for monitoring technology use.

Audits, reporting, and end-use monitoring mechanisms serve as verification tools to ensure that foreign end-users comply with requirements agreed upon in government-to-government transfers or imposed by exporting governments for direct commercial sales.[20] These mechanisms verify that defense articles, defense services, and controlled dual-use items are used only for the purposes for which they were furnished and that such items are not transferred to unauthorized parties.[21] Effective end-use monitoring requires ongoing access to foreign sites, cooperation from recipient governments, and the capacity to detect diversions or misuse after the initial transfer has occurred. For instance, the United States Department of State, through its Office of Regional Security and Arms Transfers, conducts such monitoring for defense articles, while the Department of Commerce maintains limited post-export verification programs for dual-use items.[22]

B. The Chronological Evolution of Export Controls

The history of dual-use technological regulations reflects a broader narrative arc: a shift from multilateral, cooperative frameworks in the 20th century toward increasingly fragmented, nationalized regulatory schemes driven by great power competition and technological advancement that outpaces regulatory consensus.

The foundational multilateral export control regime emerged from the geopolitical imperatives of the early Cold War. Created in 1949, the Coordinating Committee for Multilateral Export Controls (CoCom) represented the first systematic attempt by Western nations to prevent the export of strategically significant technology to the Soviet Bloc.[23] CoCom operated on a straightforward premise: if the West maintained technological superiority in military-relevant domains, that advantage could be preserved by restricting the flow of critical goods and knowledge to adversaries. Member states—primarily NATO allies plus Japan—coordinated through informal arrangements to

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[20] End-Use Monitoring of U.S.-Origin Defense Articles, U.S. DEP'T OF STATE, https://www.state.gov/end-use-monitoring-of-u-s-origin-defense-articles.
[21] *Id.*
[22] Christina L. Arabia & Michael J. Vassalotti, Cong. Rsch. Serv., R46337, *Transfer of Defense Articles: Sale and Export of U.S.-Made Arms to Foreign Entities* (Mar. 23, 2023), https://www.congress.gov/crs-product/R46337.
[23] John H. Henshaw, *The Origins of COCOM: Lessons for Contemporary Proliferation Control Regimes* 2 (The Henry L. Stimson Center, May 1993), https://www.stimson.org/wp-content/files/file-attachments/Report7_1.pdf.

control exports of weapons, dual-use technologies, and strategic materials. The regime functioned through consensus among members, who agreed to implement common export restrictions through their national legal systems.

CoCom dissolved with the Soviet Union's collapse, its *raison d'être* seemingly having been eliminated by the end of the Cold War. Yet, the perceived need for export controls did not dissipate: in 1996, 17 former CoCom members established the Wassenaar Arrangement on Export Controls for Conventional Arms and Dual-Use Goods and Technologies.[24] Unlike its predecessor, the Wassenaar Arrangement adopted a more neutral mandate, emphasizing "transparency and responsibility" in transfers of arms and dual-use goods rather than explicit containment of a defined enemy.[25] This shift denoted the optimistic post-Cold War consensus that export controls should serve multilateral non-proliferation goals as opposed to unilateral strategic advantage.

The Wassenaar Arrangement did not operate in isolation. A lattice of multilateral regimes existed in parallel to address specific technological domains: the Nuclear Suppliers Group (NSG), established in 1975, regulates nuclear exports and seeks to prevent proliferation of nuclear weapons;[26] the Australia Group, established in 1985, coordinates export controls on chemical and biological weapon precursors and related equipment;[27] and the Missile Technology Control Regime (MTCR), established in 1987, governs missiles and unmanned aerial vehicles capable of delivering weapons of mass destruction.[28] Taken together, these regimes created extensive control lists numbering in the hundreds of controlled categories—from machine tools to pathogens, from encryption software to advanced materials—which became the basis for national dual-use export legislation around the world. However, two fundamental flaws have become increasingly apparent as technological development has accelerated and geopolitical consensus has fractured.

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[24] Christopher A. Casey, *Export Controls—International Coordination: Issues for Congress*, Cong. Rsch. Serv. R47684 (Sept. 8, 2023), https://www.congress.gov/crs-product/R47684.
[25] World Trade Organization, *International Export Regulations and Controls* 123 (2023) (Part III, Chap. 5), https://www.wto.org/english/res_e/booksp_e/int_exp_regs_part3_5_e.pdf.
[26] Arms Control Association, *Nuclear Suppliers Group (NSG) at a Glance*, https://www.armscontrol.org/factsheets/nuclear-suppliers-group-nsg-glance.
[27] Center for Arms Control & Non-Proliferation, *The Australia Group: At a Glance* (Mar. 2023), https://armscontrolcenter.org/wp-content/uploads/2023/03/Australia-Group-Fact-Sheet.pdf.
[28] Nuclear Threat Initiative, *Missile Technology Control Regime (MTCR)*, https://www.nti.org/education-center/treaties-and-regimes/missile-technology-control-regime-mtcr/.

First, these multilateral bodies relied upon national implementation, voluntary adoption, informal consensus, and uneven burden sharing. None of these bodies possessed independent enforcement powers or supranational authority to compel compliance. Participation remained voluntary, and the regimes could only succeed to the extent that member states prioritized collective action over individual strategic or commercial interests. The control lists themselves, while extensive, remained limited in scope and membership: major technology producers like China and India are not members of all three regimes, creating significant gaps in global coverage. Updates to control lists required consensus among all members, a cumbersome process that often lagged behind technological development. Second, these regimes proved vulnerable to regulatory arbitrage and geopolitical fragmentation. For example, the Wassenaar Arrangement had stalled since 2022 due to deepening geopolitical rifts owing to Russia's invasion of Ukraine, which has transformed the ostensible neutral multilateral forum into a venue for great power competition.[29] Moscow began vetoing any additions to Wassenaar's control lists, blocking consensus on new controls for two years and effectively paralyzing the regime's ability to adapt to emerging technologies.[30] This obstruction incentivized discussion of creating a "Wassenaar Arrangement Minus One," a parallel framework excluding Russia, thus formalizing the very fragmentation that multilateral regimes were designed to prevent.[31] As geopolitical fractures have widened, many dual-use controls have been nationalized, leading to aggressive unilateral or regional controls of dual-use technology implemented outside multilateral frameworks.

C. Contemporary National Export Control Regimes

The frustration with multilateral gridlock has driven countries housing major AI developers to develop dedicated national legislation pertaining to export controls. The formation and maintenance of these policies signify the distinct geopolitical priorities and strategic positions of leading technology powers, creating an increasingly fragmented

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[29] Andy Bounds, *EU Consolidates Export Control Powers to Bypass Russia*, Fin. Times (Sept. 16, 2025), https://www.ft.com/content/6b0648b4-3e7c-41d1-b96a-3da2b41e4fd3.

[30] *Id.*

[31] David Shepardson & Alexandra Alper, *US, Allies Explore Russia's Exclusion from Wassenaar Export Control Regime*, The Journal of Export Controls & Sanctions (Dec. 5, 2024), https://www.worldecr.com/news/us-allies-explore-russias-exclusion-from-wassenaar-export-control-regime/.

regulatory landscape. This Note focuses on three: the United States, the European Union, and China.

1. The United States

The United States has leveraged its position as both a leading AI developer and the home of critical semiconductor technology to implement aggressive unilateral controls. The Bureau of Industry and Security (BIS), operating under the Export Administration Regulations (EAR), moved decisively in 2022 and 2023 to restrict China's access to advanced semiconductor and AI-related items.[32] These restrictions targeted not only finished chips but also the equipment necessary to manufacture cutting-edge processors, aiming to constrain China's ability to develop native advanced semiconductor capacity. The BIS classified high-end chips under the Commerce Control List—particularly categories 3A090 and 4A090 for advanced graphics processing units (GPUs)—creating licensing requirements for exports to China and other restricted areas.[33] The controls extended beyond hardware to encompass semiconductor manufacturing equipment, including lithography systems, deposition tools, and advanced packaging technology. In its final days, the Biden-Harris Administration attempted to extend these controls further into the software domain through the AI Diffusion Rule of 2024, which would have required licenses for transferring certain "advanced closed-weight AI models" to specified countries of concern.[34] The rule represented an ambitious effort to treat trained AI models as controlled items akin to physical technology. However, the rule faced immediate and intense opposition from industry stakeholders like Microsoft and Oracle who argued it would undermine American AI competitiveness, create legal uncertainty, and prove practically unenforceable given the digital nature of model

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[32] Gregory C. Allen, *Understanding the Biden Administration's Updated Export Controls*, Ctr. for Strategic & Int'l Stud. (Dec. 11, 2024), https://www.csis.org/analysis/understanding-biden-administrations-updated-export-controls.

[33] William Alan Reinsch, Matthew Schleich & Thibault Denamiel, *Insight into the U.S. Semiconductor Export Controls Update*, Ctr. for Strategic & Int'l Stud. (Oct. 20, 2023), https://www.csis.org/analysis/insight-us-semiconductor-export-controls-update.

[34] BUREAU OF INDUS. & SEC., U.S. DEP'T OF COM., *Biden-Harris Administration Announces Regulatory Framework for the Responsible Diffusion of Advanced Artificial Intelligence Technology* (Jan. 13, 2025), https://www.bis.gov/press-release/biden-harris-administration-announces-regulatory-framework-responsible-diffusion-advanced-artificial.

weights.[35] At the outset of his second term, President Trump rescinded the rule, citing these industry concerns and signaling a preference for hardware-focused controls over restrictions on AI models themselves.[36] Complementing its export restrictions, the United States has pursued a strategy of domestic capacity building through substantial subsidies. The CHIPS and Science Act of 2022 allocated approximately $52 billion in subsidies and tax credits to incentivize domestic semiconductor manufacturing, research, and workforce development.[37] This two-pronged approach—restricting adversaries' access to advanced chips while subsidizing domestic and allied production capacity—aims to maintain long-term American technological leadership rather than merely slowing competitors' progress.

2. The European Union

The European Union has pursued a parallel but distinct regulatory strategy that balances security concerns with human rights considerations and the economic interests of member states. The EU 2021/821 Dual-Use Regulation updated earlier frameworks to modernize controls on emerging technologies and explicitly address human rights impacts on dual-use exports, especially as it relates to cyber-surveillance, terrorism, and violations of international humanitarian law.[38] The update aimed to harmonize member states' rules while preserving flexibility for individual countries to invoke ad hoc controls on unlisted technologies for security or human rights reasons.[39] Despite harmonization efforts, European practice has been characterized by significant fragmentation as member states pursued unilateral measures, manifesting their distinct strategic positions in global technology supply chains. France, Spain, and the United Kingdom have unilaterally added export control requirements for certain quantum computing and semiconductor items,

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[35] Liza Lin, Corrie Driebusch & Amrith Ramkumar, *U.S. to Overhaul Curbs on AI Chip Exports After Industry Backlash*, Wall St. J. (May 7, 2025), https://www.wsj.com/tech/ai-chip-exports-overhaul-0518a821.

[36] BUREAU OF INDUS. & SEC., U.S. DEP'T OF COM., *Department of Commerce Announces Rescission of Biden-Era Artificial Intelligence Diffusion Rule, Strengthens Chip-Related Export Controls*, (May 13, 2025), https://www.bis.gov/press-release/department-commerce-announces-rescission-biden-era-artificial-intelligence-diffusion-rule-strengthens.

[37] H.R. 4346, 117th Cong. (2022) (enacted).

[38] Regulation (EU) 2021/821 of the European Parliament and of the Council of 20 May 2021 setting up a Union regime for the control of exports, brokering, technical assistance, transit and transfer of dual-use items, 2021 O.J. (L 206) 1, https://eur-lex.europa.eu/eli/reg/2021/821/oj/eng.

[39] EUR. COMM'N, *Commission Publishes Guidelines for Cyber-Surveillance Exporters* (Oct. 16, 2024), https://policy.trade.ec.europa.eu/news/commission-publishes-guidelines-cyber-surveillance-exporters-2024-10-16_en.

going beyond the EU's common control list.[40] The Netherlands, as the home of ASML, occupies a uniquely powerful position in the semiconductor supply chain. Recognizing this leverage, the Netherlands formed a trilateral agreement with the United States and Japan to curb chip exports to China, effectively implementing controls negotiated outside the EU framework, albeit with Brussels' tacit support.[41]

3. China

China has responded to Western export controls with a deliberate strategy of building reciprocal regulatory capacity and leveraging its own technological strengths as bargaining chips. The 2020 Export Control Law represented China's first comprehensive national legislation restricting exports of dual-use items and military goods for the "maintenance of national security and national interests"—language sufficiently broad to encompass any technology Beijing deems strategically significant.[42]

Initially, China's export controls appeared more symbolic than substantive, as China remained primarily a technology importer rather than exporter in the most advanced domains. However, China's regulatory posture has evolved as its technological capabilities have matured and as Beijing has recognized leverage points in its own supply chains. By December 2024, China further tightened restrictions with the Regulations on Export Control of Dual-Use Items Law, consolidating various control lists covering nuclear, biological, chemical, and missile-related items while explicitly expanding oversight to high-tech sectors, including semiconductors and AI.[43] Crucially, China's export controls have begun to target the same domains in which it faces import restrictions: semiconductors and AI algorithms. In doing so, Beijing signals that it will not only comply

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[40] Jasper Helder et al., *International Unilateral Export Controls—An Increased Focus on Advanced Technologies*, Akin Gump Strauss Hauer & Feld LLP (Apr. 5, 2024), https://www.akingump.com/en/insights/alerts/international-unilateral-export-controls-an-increased-focus-on-advanced-technologies.

[41] Gregory C. Allen, Emily Benson & Margot Putnam, *Japan and the Netherlands Announce Plans for New Export Controls on Semiconductor Equipment*, Ctr. for Strategic and Intl. Stud. (Apr. 10, 2023), https://www.csis.org/analysis/japan-and-netherlands-announce-plans-new-export-controls-semiconductor-equipment.

[42] *Export Control Law of the People's Republic of China* (promulgated Oct. 17, 2020, effective Dec. 1, 2020), http://www.npc.gov.cn/englishnpc/c2759/c23934/202112/t20211209_384804.html.

[43] *China Promulgates New Dual-Use Export Control Regulations*, Covington & Burling LLP (Oct. 25, 2024), https://www.cov.com/en/news-and-insights/insights/2024/10/china-promulgates-new-dual-use-export-control-regulations.

with global non-proliferation norms but also leverage export controls to guard its own technological advantages and impose costs on countries restricting Chinese market access.

This strategic mirror became particularly evident in China's October 2025 controls on rare earth elements, advanced battery materials, and gallium and germanium—materials essential for semiconductor production, in which China holds dominant market positions.[44] China has also announced plans for an algorithm regulation regime and maintains an "unreliable entities" list to sanction foreign technology companies perceived as threatening Chinese interests. This tit-for-tat approach demonstrates that export controls, once primarily a tool of Western powers seeking to maintain technological advantage over rivals, have become weapons in a broader technology competition where multiple powers possess leverage in different segments of complex global supply chains.

II. Enforcement Gaps in AI

A. Defining the Enforcement Gap

Modern export control regimes are animated by a foundational, durable assumption: that strategically sensitive technologies can be identified, regulated, and enforced through discrete acts of transfer. Whether the controlled object is a physical artifact, a component, or a bundle of technical information, the law presumes that national security risk materializes the moment an item changes hands or crosses a jurisdictional boundary, thereby generating an "export event" that can be licensed, interdicted, or penalized. This assumption has historically aligned well with the regulation of tangible dual-use goods, such as drones, chemicals, and machine parts, whose ownership can be traced to identifiable legal entities, and whose movement can be observed at customs chokepoints that function as practical enforcement bottlenecks. Artificial intelligence disrupts this enforcement logic not by presenting a marginally harder case, but by invalidating the very premises on which export control law has been constructed. It requires a reconciliation of the existing legal framework optimized for regulating movement with a technological ecosystem in which capability is increasingly conferred through access, use, and recombination rather than transfer.

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[44] Gracelin Baskaran, *China's New Rare Earth and Magnet Restrictions Threaten U.S. Defense Supply Chains*, Ctr. for Strategic & Int'l Stud. (Oct. 9, 2025), https://www.csis.org/analysis/chinas-new-rare-earth-and-magnet-restrictions-threaten-us-defense-supply-chains.

The export control paradigm presupposes that controlled technologies are rivalrous and excludable, such that restricting possession meaningfully constrains downstream use. Advanced AI systems, by contrast, derive their strategic value from inputs that are neither co-located nor transferred as a unified object. Compute, model weights, and training pipelines—each individually indispensable to frontier AI capability—are now routinely decoupled across multiple jurisdictions, firms, and layers of abstraction. A foreign entity does not need to own a data center or acquire a single physical artifact to train or deploy advanced models; it only needs to obtain the ability to invoke large-scale computation, often through infrastructure that remains physically located within the United States or allied jurisdictions. In such cases, no export occurs in the traditional sense, even as the functional outcome is indistinguishable from the foreign acquisition of export-controlled AI hardware.

In practice, this gap widened through the service-based reorganization of the AI stack, decades in the making. Where earlier generations of sensitive dual-use technology were sold as products, advanced AI compute is now predominantly offered as an on-demand service, provisioned remotely through cloud platforms that act as an intermediary to vast concentrations of specialized hardware. Export controls that successfully restrict the sale of GPUs or accelerators to certain jurisdictions therefore operate only on one axis of a much broader capability surface if the computational power embodied in those same chips remains available for purchase as a service. Only with higher-capacity licensing, monitoring, and end-use restrictions can the legal regime effectively uphold the intention of American export controls.

B. The Hardware-Computation Mismatch of AI

The contemporary emphasis on hardware export controls as the primary level of AI governance rests on yet another intuitively appealing but incomplete premise: that chips, accelerators, and interconnects are the most traceable elements of the AI stack from a regulatory perspective. Unlike data or algorithms, advanced semiconductors are physical, serial-numbered, logistically traceable, and produced through a highly concentrated supply chain that affords regulators meaningful points of intervention. These features have made hardware controls the centerpiece of recent U.S. export policy, particularly with respect to advanced GPUs used in large-scale model training. However, this focus obscures a critical

distinction between hardware as an object and computation as a function. Hardware enables compute, but compute itself is not a product that is bought, shipped, or owned in the traditional sense. Export controls that operate exclusively on the sale or shipment of chips therefore regulate only the physical substrate of AI capability, not the computational capacity that it makes available.

Under existing law, the provision of compute-as-a-service occupies an ambiguous and largely unregulated space. An "export" under EAR § 734.14 requires the shipment or transmission of items subject to the EAR out of the United States—a condition that is not satisfied when advanced chips remain physically located in domestic data centers.[45] Similarly, an "export of technology" under EAR § 734.15 requires the release of controlled technology to a foreign national through mechanisms such as visual inspection or application of the technology abroad.[46] Providing compute services typically involves none of these acts: the foreign user receives the results of computation, not the underlying chip designs, specifications, or manufacturing know-how that the EAR traditionally treats as controlled technology. As a result, large-scale access to U.S. compute infrastructure by foreign entities often falls entirely outside the doctrinal scope of export controls.

This hardware-computation mismatch becomes further pronounced when considering software and model diffusion. Modern AI architecture is built upon a foundation of openly published research intentionally placed in the public domain, consistent with well-established scientific norms. The transformer architecture, first articulated in the "Attention Is All You Need" paper in 2017, rapidly became the default paradigm for natural language processing and now underlies virtually all large language models.[47] Attempting to retroactively restrict the dissemination of such foundational research is both futile and harmful to the development of AI from the computer science community.

Model weights further complicate this picture. Unlike blueprints or schematics, weights are precise numerical parameters that define the behavior and output of an AI model, not intelligible design documents nor physical objects. Once released—whether through open-source licensing, leaks, or reverse engineering—weights can be fine-tuned,

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[45] 15 C.F.R. § 734.14 (1996).

[46] 15 C.F.R. § 734.15 (1996).

[47] Ashish Vaswani et al., *Attention Is All You Need*, 31 Advances in Neural Info. Processing Sys. 5998 (2017), https://arxiv.org/abs/1706.03762.

repurposed, and deployed by actors for drastically different purposes.[48] These characteristics place model weights in an uneasy position relative to existing export control categories. They are not "technology" as the BIS has traditionally defined it, nor are they tangible items that can be inventoried or seized. Once weights are made public, they cannot be unreleased, a fact that has led to the proposal for ex-ante evaluation and governance of open-weight releases as the only viable point of intervention.[49]

Faced with this reality, U.S. policymakers face a limited and unsettled set of options.[50] One approach would seek to restrict the availability of open model weights altogether, treating them as controlled artifacts despite their incompatibility with existing definitions. Another strategy would be to invest in the institutional capacity to continuously evaluate the dual-use AI ecosystem, enabling regulators to respond dynamically to emerging risks without relying on blanket prohibitions. A third would be to accept, or even affirmatively promote, openness as a strategic choice, prioritizing diffusion and defensive advantage over attempts at containment. To date, policy has oscillated uneasily among these approaches. The now-rescinded AI Diffusion Rule represented a short-lived yet ambitious effort to move toward direct control of model distribution; AI safety institutes and public-private partnerships gesture toward ecosystem-level evaluation without clear mandates or enforcement authority; and the research community continues to pursue openness largely independent of export control logic. The most strategically salient components of the technology do not map cleanly onto the categories the law is equipped to regulate. Hardware controls address objects, not functions. Technology controls address disclosures, not capabilities. Software controls struggle against the realities of open publication and digital replication. Until these mismatches are confronted directly, AI export controls will remain simultaneously aggressive in form and incomplete in effect.

C. Export Control Evasion

Existing regulatory frameworks rest on the assumption of transparent cross-border transactions, but determined adversaries persist through opacity. For nations

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[48] NAT'L TELECOMM. & INFO. ADMIN., *Dual-Use Foundation Models with Widely Available Model Weights* (July 2024), https://www.ntia.gov/sites/default/files/publications/ntia-ai-open-model-report.pdf.
[49] *Id.*
[50] *Id.*

subject to stringent import restrictions, particularly China, the incentive structure shifts decisively: rather than competing within the regulated market, state-sponsored entities and private firms alike pursue alternative pathways to acquire controlled technologies while circumventing detection. These evasion tactics can broadly be categorized into three buckets: shell companies and front organizations, cloud compute laundering, and AI model distillation and reverse engineering.

The most direct evasion strategy leverages corporate form itself to obscure intention. Shell companies and front organizations allow restricted end-users to disguise their identity, nationality, or institutional affiliation by routing transactions through legally separate entities registered in jurisdictions with minimal export control enforcement or weak beneficial ownership disclosure requirements.[51] These intermediaries function as pass-through buyers, acquiring controlled items ostensibly for legitimate commercial purposes within an unrestricted jurisdiction. Then, items are redirected to the actual end-user through secondary transactions that fall outside the original exporting nation's regulatory visibility. This practice escalated notably through 2024 and into 2025 as U.S. semiconductor export controls tightened.[52] Reporting from June 2025 documented Chinese firms systematically accessing U.S.-manufactured AI chips through entities registered in Singapore and Malaysia, exploiting these jurisdictions' less-stringent end-use verification protocols and their position as regional technology distribution hubs.[53] Four months later, Singaporean company Megaspeed was alleged to have routed nearly $2 billion worth of NVIDIA chips to China.[54]

The shift toward cloud-based AI development creates a second, more insidious evasion pathway that operates entirely within the service layer. Cloud compute laundering exploits the ambiguity identified earlier: that providing remote access to controlled hardware does not constitute an "export" under existing regulatory definitions, even when the functional outcome is no different than physical delivery of that hardware to a foreign

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[51] BUREAU OF INDUS. & SEC., U.S. DEP'T OF COM., *Don't Let This Happen to You!* 15–18 (July 2024), https://www.bis.gov/media/documents/dlthty7-1-2024.pdf.

[52] Karen M. Sutter, *U.S. Export Controls and China: Advanced Semiconductors* 8–10 (Cong. Rsch. Serv. Rep. No. R48642, Sept. 19, 2025), https://www.congress.gov/crs-product/R48642.

[53] Ying Xian Wong, *Malaysia Probing Reports of Chinese Firms' Use of Nvidia AI Chips*, *Wall St. J.* (June 18, 2025), https://www.wsj.com/tech/ai/malaysia-probing-reports-of-chinese-firms-use-of-nvidia-ai-chips-363651b9.

[54] Singapore police probe Nvidia customer Megaspeed over alleged export violations, *CNBC* (Oct. 10, 2025), https://www.cnbc.com/2025/10/10/singapore-us-investigate-nvidia-client-megaspeed-export-controls-violation.html.

adversary. Chinese entities can obtain compute capacity equivalent to thousands of restricted GPUs without ever taking possession of a single chip, simply by purchasing cloud services from providers whose data centers are physically located in the U.S. or allied jurisdictions. By late 2024, at least eleven Chinese entities had sought access to restricted U.S. technologies through cloud service subscriptions, particularly targeting Amazon Web Services, Microsoft Azure, and Google Cloud Platform.[55] These entities did not register as Chinese companies, instead operating through subsidiaries, academic partnerships, and corporate affiliates designed to obscure beneficial ownership and ultimate end-use. The cloud providers, lacking clear regulatory guidance on compute-as-a-service transactions and facing intense competitive pressure in the global market, often provisioned access based on facially legitimate business justifications without conducting deep due diligence. Unlike physical chip sales, which require export licenses and trigger customs inspections, cloud service contracts are treated as ordinary commercial agreements for computational time rather than technology transfers, falling outside the scope of the EAR.

The third evasion pathway operates at the knowledge layer, exploiting the intangible and replicable nature of trained AI models. Model distillation refers to a machine learning technique in which a smaller, computationally cheaper "student" model is trained to replicate the behavior of a larger, more capable "teacher" model by learning from the teacher's outputs rather than the original training data.[56] When applied as an evasion strategy, distillation allows adversaries to extract the functional capabilities of restricted frontier models without obtaining the model weights themselves or the computational infrastructure used to train them. The DeepSeek controversy in January 2025 is illustrative. Chinese AI laboratory DeepSeek released a reasoning model that exhibited performance characteristics strikingly similar to OpenAI's o1 model, despite China's limited access to advanced AI accelerators under U.S. export controls.[57] A deeper analysis of DeepSeek's training methodology suggested the distillation of OpenAI's model

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[55] Eduardo Baptista, Fanny Potkin & Karen Freifeld, *Chinese Entities Turn to Amazon Cloud, Its Rivals, to Access High-End U.S. Chips, AI*, Reuters (Aug. 22, 2024), https://www.reuters.com/technology/chinese-entities-turn-amazon-cloud-its-rivals-access-high-end-us-chips-ai-2024-08-23/.

[56] Geoffrey Hinton, Oriol Vinyals & Jeff Dean, *Distilling the Knowledge in a Neural Network*, NIPS 2014 Deep Learning Workshop (2014), https://arxiv.org/abs/1503.02531.

[57] Laurie Harris, *DeepSeek and the Race to Develop Artificial Intelligence*, Cong. Rsch. Serv., In Focus No. IF13051 (July 1, 2025), https://www.congress.gov/crs-product/IF13051.

from outputs generated by OpenAI's application programming interface (API). OpenAI subsequently accused DeepSeek of violating its terms of service and potentially engaging in intellectual property theft, though the legal basis for such claims remains contested: the outputs generated by an API are not themselves subject to export controls, and distillation from outputs occupies a gray area in both IP law and export control doctrine.[58] The distilled model may be less capable than the original, but it can achieve functionally similar performance for many applications while evading the regulatory barriers that would prevent direct acquisition of the teacher's model weights or the computational infrastructure used to create them. Distillation represents a structural challenge to export control efficacy in the AI domain. As long as frontier models are accessible through commercial APIs or open-source releases (e.g., Meta's Llama 2), adversaries can extract their capabilities through techniques that leave no regulatory footprint. The teacher model never needs to leave U.S. jurisdiction, the student model is trained using compute that falls below regulatory thresholds, and the knowledge transfer occurs. In any case, export control policy alone cannot confront the challenges posed by a post-deployed environment.

III. Proposals for New Dual-Use Technology Regulatory Schemes

A. Creating a Compute Regulation Framework

Part II mapped out the contemporary export control regime as ill-equipped to regulate frontier AI models, where capability is conferred through (1) access to large-scale compute, (2) software artifacts that replicate instantly, and (3) distributed training pipelines that evade any single export event. This Part examines a layered approach that brings American export controls back into contact with the capability surface they are supposed to govern. Specifically, it sketches a three-tier framework that relies on existing export-control mechanisms—such as list-based controls, end-user controls, end-use controls, and services controls—but reassigns them to the points where modern compute travels: through cloud provisioning, orchestration, and post-deployment scaling.

Export controls already contain the conceptual building blocks for regulating activity rather than goods. Export Control Classification Numbers (ECCNs) and the CCL structure can be iteratively updated to control new "items," including those that are not

[58] *Id.*

traditional hardware. BIS already uses an evolving mix of list-based controls, end-user controls via the Entity List, end-use controls, and services controls (including the "U.S. persons" rule).[59] Many of these tools can be retrofitted into a capability-triggered licensing regime that attaches obligations to (1) cloud compute providers and (2) operators of large compute clusters, with thresholds keyed to a mix of hardware characteristics and cluster-level indicators. There are three tiers to this framework: **object control** (what counts as a controlled "item"), **capability control** (who must do due diligence at the access layer), and **behavior control** (how misuse is detected and stopped after access is granted).

1. Tier 1: Object Control

Tier 1 makes the smallest conceptual move with the largest downstream effect: it regards virtual compute access as an exportable "item" when it functionally substitutes for the delivery of controlled hardware. The relevant transfer event is often no longer the shipment of GPUs but rather the grant of time-sliced access to GPU clusters, which constitutes the precise moment at which capability is conferred onto third parties.

The first doctrinal step requires amending the EAR to explicitly define "compute services" as items subject to export licensing requirements. When a cloud provider grants foreign nationals or foreign-located entities access to advanced AI training infrastructure, that grant should trigger the same licensing obligations as the physical sale of controlled chips. This requires updating the CCL to add new categories that explicitly cover compute provisioned as a service rather than as a product. The updated CCL should delineate three distinct service categories, each representing a different layer of the cloud computing stack. First, Infrastructure-as-a-Service (IaaS) offerings that provide direct access to advanced AI accelerators—particularly GPUs classified under ECCNs 3A090 and 4A090—should be treated as controlled items when provisioned to foreign persons or entities in countries of concern.[60] IaaS represents the more direct analogue to physical hardware transfer: the customer obtains access to bare-metal compute resources and exercises near-total control over how those resources are deployed, differing from physical ownership only in the legal title to the underlying equipment.

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[59] 15 C.F.R. §§ 744.6 (U.S. persons export control), 744.16 (Entity List), 744.21 (military end uses and users) (1996).

[60] 15 C.F.R. pt. 774, supp. 1, cat. 3 (2024) (electronics and advanced computing items).

Second, Platform-as-a-Service (PaaS) tools that enable model training should be classified as controlled items when they expose capabilities equivalent to or derived from controlled hardware. PaaS abstracts away from infrastructure management, providing users with pre-configured environments optimized for machine learning workflows. Platforms like Amazon SageMaker, Google Vertex AI, and Microsoft Azure Machine Learning offer managed services for data preprocessing, distributed training, hyperparameter optimization, and model deployment—capabilities that allow foreign entities to train frontier models without needing to understand or manage the underlying hardware architecture.[61] While PaaS introduces an additional layer of abstraction between the user and controlled chips, the functional capability it confers remains strategically equivalent to authorized hardware access.

Third, Software-as-a-Service (SaaS) applications that expose restricted algorithmic capabilities through commercial APIs should be regulated when they enable adversaries to extract or replicate the functional capabilities of frontier models. This category addresses the distillation and reverse-engineering pathways identified in Part II: when a model trained on controlled compute infrastructure is deployed through an API, systematic querying of that API can allow adversaries to distill the model's capabilities into their own systems without ever accessing the underlying weights or compute. While SaaS controls present the most challenging enforcement equations—distinguishing between legitimate commercial API usage from systemic capability extraction requires behavioral monitoring rather than point-of-transfer licensing—the regulatory gap they address is precisely the one that allows adversaries to circumvent hardware and software controls simultaneously.

Importantly, not all compute use constitutes strategic capability. Imposing export licensing requirements on consumer cloud storage, basic web hosting, or routine enterprise applications would create massive regulatory overhead while failing to address national security risks. The framework should establish computational thresholds that distinguish controlled from uncontrolled services based on objective, measurable criteria tied to the technical requirements of frontier AI development. The thresholds should be defined using a combination of metrics that collectively indicate large-scale model training capability so as not to risk chilling innovation or inducing over-compliance. Total

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[61] Dhavalkumar Patel et al., *Cloud Platforms for Developing Generative AI Solutions: A Scoping Review of Tools and Services*, arXiv:2412.06044 (Dec. 8, 2024), https://arxiv.org/abs/2412.06044.

computational capacity measured in petaFLOPs (floating-point operations per second, or FLOPs) provides a hardware-agnostic measure of aggregate processing power.[62] A threshold of 10^{25} petaFLOPs would capture compute clusters capable of training models at the frontier of current capabilities while excluding smaller-scale applications. Training run scale, measured in total floating-point operations, offers a usage-based metric that accounts for both the computational capacity and duration over which it is employed. A threshold of 10^{25} FLOPs would align with the computational budgets used to train contemporary large language models while remaining several orders of magnitude above typical commercial applications.[63]

Hardware specifications provide a third dimension for threshold determination. Chips optimized for AI workloads exhibit distinct characteristics: support for reduced-precision formats like FP16 (16-bit floating point) and INT8 (8-bit integer) arithmetic, high-bandwidth memory architectures, and specialized tensor processing units designed specifically for matrix operations central to neural network training. Services provisioning access to hardware with these specifications at scale should trigger heightened scrutiny regardless of the nominal purpose for which the compute is requested. The combination of these metrics—capacity, usage intensity, and hardware specialization—creates a multidimensional threshold that adapts to technological advancement while remaining administratively feasible. Notably, these thresholds must be dynamic rather than static. The computational requirements for frontier AI capabilities have increased by roughly an order of magnitude annually over the past decade, a trend that shows no sign of slowing down.[64] Hardcoding specific numerical thresholds into regulation would create brittleness: what constitutes frontier capability today may become commodity capability within two years in light of exponential improvements in compute efficiency. The regulatory framework should delegate threshold-setting authority to BIS with a mandate to review and update thresholds annually based on the current state of AI development, ensuring that controls track capability rather than lag behind on technological progress.

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[62] Matteo Pistillo, Suzanne Van Arsdale, et al., *The Role of Compute Thresholds for AI Governance*, 1 Geo. Wash. J. L. & Tech 26 (2025).

[63] *Id.*

[64] Jaime Sevilla et al., *Compute Trends Across Three Eras of Machine Learning*, arXiv:2202.05924 (Mar. 9, 2022), https://arxiv.org/abs/2202.05924.

Traditional export controls place compliance burdens on exporters and end-users: the entity seeking to export controlled items must obtain a license, and the end-user must not be listed on the Entry List of engaged in prohibited end-uses. This model assumes that the exporter has visibility into the transaction and sufficient control over the technology to prevent unauthorized transfers. Cloud computing inverts this structure: the end-user may be several layers removed from the entity that owns the controlled hardware, separated by resellers, system integrators, and subcontractors. The cloud provider, not the end-user, possesses actual control over whether access is granted or terminated. Accordingly, licensing obligations must be imposed directly on cloud providers rather than relying on end-users to self-report. Before activating services that exceed the computational thresholds identified above for customers in restricted jurisdictions or for customers that cannot verify their identity, nationality, or organizational affiliation, providers should be required to submit licensing applications to BIS.[65] This front-loads compliance, as the provider cannot provision the service until the license is granted, preventing compute laundering.

The licensing process itself should mirror existing EAR procedures while accounting for the unique characteristics of service provisioning. Applications should identify the customer entity, beneficial ownership structure, stated end-use purpose, requested computational capacity, and anticipated duration of access. For customers in countries of concern or customers affiliated with entities on the Entity List, the presumption should be denial unless the applicant demonstrates that the compute will not be used for purposes that threaten U.S. national security—a burden-shifting mechanism that accounts for the heightened risk profile that these transactions present.[66]

Complementing the licensing requirement, providers should maintain robust record-keeping systems that enable post hoc audits and violation procedures, computational resources provisioned, actual usage patterns, and any anomalies flagged by automated monitoring systems. The records should be retained for a minimum of five years and made available to BIS upon request, creating an audit trail that allows regulators to detect patterns of abuse, identify shell company networks, and trace compute access back to ultimate beneficial users even when transactions are obscured through

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[65] 15 C.F.R. § 748.1 (2026) (establishing general export licensing procedures).
[66] 15 C.F.R. § 744.21(e).

intermediaries.[67] The record-keeping mandate transforms cloud providers into de facto gatekeepers, a role they already perform for anti-money laundering and counter-terrorism financing purposes in the financial services context.

Compute services do not respect national boundaries. A customer in Beijing can contract with a cloud provider headquartered in Seattle, which provides access to compute infrastructure physically located in Virginia, using chips manufactured in Taiwan from designs created in California, all without any physical object crossing a border. This jurisdictional complexity creates enforcement challenges that threaten to undermine the entire regulatory framework if unaddressed.

There are several challenges that confront U.S. export controls in the AI context. The first is the extraterritorial application of U.S. export controls. The EAR already asserts jurisdiction over foreign-made items that incorporate more than a de minimis amount of U.S.-origin content, a principle prominently deployed in the "Foreign Direct Product Rule" that restricts exports of semiconductors manufactured abroad using U.S. technology.[68] Extending this principle to cloud services would assert U.S. jurisdiction over any compute provisioned using chips designed or manufactured with U.S. technology, regardless of where the data center is physically located or where the provider is incorporated. While this maximalist approach would close any jurisdictional gaps, it risks provoking international backlash and may exceed the statutory authority granted under the Export Control Reform Act (ECRA) of 2018. A more measured approach would focus on services provided by U.S. persons or using U.S.-origin hardware, grounding jurisdiction in traditional territorial and nationality principles. Cloud providers incorporated in the United States or employing U.S. persons in roles that involve providing controlled compute would be subject to the licensing requirements regardless of where their infrastructure is located.[69] Similarly, services utilizing hardware physically located in the United States or chips classified under controlled ECCNs would trigger licensing obligations regardless of the provider's nationality. This approach maintains regulatory reach while remaining rooted in firmer legal foundations.

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[67] 15 C.F.R. pt. 762 (1996) (requiring recordkeeping for export transactions).
[68] 15 C.F.R. § 734.9 (1996) (Foreign Direct Product Rule).
[69] 15 C.F.R. § 744.6.

The second challenge is regulatory fragmentation across jurisdictions. Unilateral U.S. controls on compute services will prove effective only if allied nations adopt comparable frameworks; else, adversaries will simply route their cloud computing purchases through providers in jurisdictions with lax enforcement. The Wassenaar Arrangement, despite its recent paralysis discussed in Part I, provides an existing institutional structure for multilateral harmonization of dual-use export controls. A more expansive plurilateral agreement modeled on the Netherlands-Japan-U.S. trilateral arrangement on semiconductor manufacturing equipment could help harmonize compute service controls among the jurisdictions that host the major cloud providers and chip manufacturers, establishing common definitions of controlled compute services, standardized computational thresholds, and reciprocal enforcement mechanisms that preclude regulatory arbitrage. The agreement should include information-sharing provisions that allow participating states to trace compute access across jurisdictional boundaries, identify attempts at regulatory evasion through provider-shopping, and coordinate enforcement actions against systematic violators.

2. Tier 2: Capability Control

Tier 1 establishes what items are controlled and under what circumstances licenses are required. Tier 2 addresses how providers verify customer identity, assess end-use risk, and monitor ongoing compliance—shifting the compliance burden from discrete transfer points to continuous access points. Where traditional export controls operate as gates that close at the moment of shipment, capability controls operate as ongoing obligations that persist through the service relationship.

Customer identity and beneficial ownership structure verification is not without precedent. The Bank Secrecy Act of 1970 marked the first move to curtail money laundering, terrorist financing, and sanctions evasion.[70] Most recently, the Customer Identification Program Rule of 2010[71] and the Customer Due Diligence Rule of 2016 imposed detailed know-your-customer (KYC) obligations, requiring banks to collect

[70] 12 U.S.C. § 1951 note (2018).

[71] 31 C.F.R. § 1020.220 (2010) (Customer Identification Program requirements for banks).

identifying information, verify that information through documentary and non-documentary methods, and maintain records of the verification process.[72]

Cloud providers offering controlled compute services should face analogous obligations. Before provisioning services that exceed the computational thresholds established in Tier 1, providers should verify the customer's legal identity, nationality, and organizational affiliation through reliable, independent documentation.[73] For individual customers, this includes government-issued identification, nationality verification, and screening against the Entity List and Denied Persons List databases. For corporate customers, providers must identify beneficial owners—the natural persons who ultimately own or control the entity—and verify that none of those persons are subject to export restrictions. The verification process must extend beyond the facial review of submitted documentation. Providers should be required to cross-reference customer information across multiple data sources: corporate registries, sanctions lists, trade databases, and adverse media reports. For customers located in jurisdictions with weak beneficial ownership disclosure requirements or those operating in sectors known to be vectors of export control evasion, providers should conduct due diligence, including source-of-funds verification and purpose-of-business inquiries.[74] KYC requirements also create opportunities for pattern detection. When multiple apparently unrelated customers share beneficial owners, registration addresses, or payment sources, this clustering suggests coordinated evasion activity. When a customer's stated business purpose—a regional e-commerce platform, for instance—bears no relationship to the computational capacity requested, this discrepancy warrants closer inspection. Providers should be required to maintain systems that flag these anomalies and escalate them for human review, creating a first line of defense against the shell company and cloud laundering strategies mapped in Part II.

An additional consideration for capability control is telemetry. If KYC requirements address who the customer is, telemetry requirements address what the customer does with the provisioned compute. Telemetry refers to the automated

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[72] Customer Due Diligence (CDD) Final Rule, Fin. Crimes Enf't Network, https://www.fincen.gov/resources/statutes-and-regulations/cdd-final-rule (last visited Jan. 27, 2026).
[73] *Id.* § 1020.220(a)(2).
[74] Fin. Action Task Force, *Guidance on Beneficial Ownership of Legal Persons* (Mar. 2023), https://www.fatf-gafi.org/content/dam/fatf-gafi/guidance/Guidance-Beneficial-Ownership-Legal-Persons.pdf.

collection and transmission of usage data from remote systems, allowing providers and regulators to monitor computational activity in near-real-time without requiring physical access to the infrastructure or direct inspection of the customer's workloads.[75] Cloud providers already collect extensive telemetry for billing, capacity planning, and service optimization purposes.[76] This telemetry captures metadata about computational activity: where resources were accessed, for how long, and with what network traffic profiles. Crucially, this metadata can be collected without accessing the content of the customer's data or the specifics of their model—partially addressing higher-order privacy concerns that would arise if monitoring required providers to inspect training data or model architectures.[77] For customers accessing controlled compute services, providers should be required to collect and retain enhanced telemetry that enables detection of frontier AI training workloads. Key indicators include sustained high-utilization of GPU clusters over extended periods (measured in days or weeks rather than hours), memory access patterns characteristic of large-scale neural network training, inter-node communication volumes indicative of distributed training across multiple machines, and checkpoint writing behavior that suggests iterative model development. These patterns collectively distinguish frontier model training from other compute-intensive applications like rendering, simulation, or data analytics.

The telemetry data serves two distinct regulatory functions. First, it allows providers to verify that customers are using compute consistent with their stated end-use purposes. For instance, a customer who claimed to need capacity for scientific simulation but exhibits usage patterns demonstrative of large language model training has misrepresented their intentions and should face service suspension pending investigation. Second, aggregated telemetry enables BIS to monitor compliance at an ecosystem level, identifying trends in how controlled compute is being used, which jurisdictions are consuming the most frontier-capable resources, and whether usage patterns suggest systematic evasion across multiple providers or customer accounts.

Beyond verifying identity and monitoring usage, providers must assess the substantive purpose for which customers seek controlled compute. End-use assessment

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[75] Nick Gallagher & Michael Goodwin, *Telemetry*, Int'l Bus. Machs. Corp. (Oct. 13, 2025), https://www.ibm.com/think/topics/telemetry.
[76] *Id.*
[77] Sastry et al., *supra* note 9, at 40–42.

requires customers to disclose their intended applications in detail sufficient to evaluate dual-use risks before services are activated. This disclosure creates a baseline against which actual usage can be measured and establishes a contractual representation that can support enforcement actions if violated.

The assessment should flag several categories of risk. Vague or generic descriptions—"AI research," "data analysis," or "technology development"—provide insufficient basis to evaluate dual-use concerns and should prompt requests for additional specificity.[78] Requests for computational capacity that vastly exceed what the customer's state needs would require may suggest misunderstanding at best, or misrepresentation of purpose or plans to scale beyond disclosed applications at worst.

On a contractual level, service agreements should explicitly prohibit customers from using controlled compute for military applications, weapons development, mass surveillance systems, or other applications that would violate export control end-use restrictions. While this occurs at a surface level through acceptable use policies, the deficit of capability control prevents companies from meaningfully enforcing these policies. Contractual provisions with improved end-use monitoring create private law obligations that supplement public law export controls: even if a particular use might fall outside the technical scope of export restrictions, a customer who violates contractual use limitations has committed breach of contract, enabling the provider to terminate service and supporting civil enforcement actions.

3. Tier 3: Behavior Control

Tiers 1 and 2 operate at the threshold of access: they determine who receives controlled compute and under what conditions. Tier 3 addresses what happens after access is granted, recognizing that some customers who pass identity verification and end-use assessment may subsequently divert compute to prohibited applications. Behavior control mechanisms enable detection and remediation of misuse through post-deployment monitoring and intervention.

The most aggressive behavior control mechanism is the ability to remotely terminate access to controlled compute when misuse is detected or suspected. Remote deactivation—colloquially referred to as a "kill switch"—allows providers to instantly cut

[78] 15 C.F.R. § 748.4 (2025).

off a customer's access to computational resources, preventing continued misuse while investigations proceed. Implementation requires both technical and legal infrastructure.[79] Technically, cloud architecture already supports granular access controls that can disable specific customer accounts or workloads without affecting other users of the same physical infrastructure. Providers frequently employ these controls to address non-payment, terms-of-service violations, and security incidents. Extending this capability to export control enforcement requires integrating automated monitoring systems that flag suspicious usage patterns with access control systems that can execute immediate termination upon provider or regulatory direction.

Legally, service agreements must explicitly reserve the provider's right to terminate access without notice when misuse is detected or when directed by government authorities pursuant to export control requirements. This reservation addresses potential contractual liability claims: customers cannot credibly argue that service termination breaches the contract when the contract itself authorizes such termination under specific circumstances. The terms should clarify that providers may retain customer data for forensic and enforcement purposes even after termination, preventing customers from destroying evidence simply by canceling their accounts.

The kill switch mechanism is not without complications. False positives—that is, terminating legitimate customers based on algorithmic misclassification of their usage patterns—impose real costs and risk undermining provider relationships with enterprise customers. Adversarial customers may also attempt to design their usage patterns to evade detection triggers or to complete training runs before termination can be executed. Despite these limitations, remote deactivation provides a critical capability that traditional export controls lack entirely: the ability to revoke access after it has been granted.

Conclusion

The stakes of artificial intelligence transcend strategic competition or national security. AI is the first general-purpose technology in human history that defies the characteristics we have come to expect from dual-use items: it is highly intangible,

[79] Gabriel Kulp et al., *Hardware-Enabled Governance Mechanisms: Developing Technical Solutions to Exempt Items Otherwise Classified Under Export Control Classification Numbers 3A090 and 4A090* (Rand Corp. WRA3056-1, Jan. 2024), https://www.rand.org/content/dam/rand/pubs/working_papers/WRA3000/WRA3056-1/RAND_WRA3056-1.pdf.

scalable, and diffusive. The decisions made now about AI governance will determine not merely which nations maintain technological advantage, but what kind of technology gets built, by whom, for what purposes, and under what constraints. These choices will shape the character of intelligence as it becomes increasingly divorced from human oversight.

Current law is wholly incongruous with AI's constitutive and allocative characteristics. Export controls designed for discrete physical transfers cannot regulate capabilities conferred through cloud access, distilled through API queries, or replicated through open-source releases. Compute remains the sole tractable chokepoint, but the window to create comprehensive governance is closing. The computational thresholds proposed in this Note are moving targets that will require constant recalibration as AI develops at breakneck speeds. The concentration of compute that makes governance possible faces pressure from open-source development, distributed training, and algorithmic efficiency improvements that threaten to dissolve the very chokepoints upon which regulation depends. Importantly, efforts to contain unauthorized AI diffusion do not stop countries from achieving improvements on their own, but they buy time that allows durable governance (i.e., multilateral AI frameworks and safety norms) to take hold. The kairos exists now, while compute remains centralized and supply chains are identifiable.